Delay-aware Decentralized Federated Learning in LEO Satellite-assisted Vehicular Networks

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***Abstract*—Low Earth orbit (LEO) satellite-assisted decentralized federated learning (DFL) has emerged as a promising solution that enables moving intelligent vehicles to train machine learning (ML) models for smart services in vehicular networks. In LEO satellite-assisted DFL, satellites train a global model by exchanging and aggregating model parameters themselves via inter-satellite links without the need for a central server. Nevertheless, as each satellite simultaneously executes computationally intensive inference tasks, the different waiting times across satellites increase the overall FL delay and degrade the accuracy of the global model. To address this, we propose a delay-aware DFL framework in LEO satellite-assisted vehicular networks, considering the waiting delay caused by simultaneously executed inference tasks. We formulate a satellite server selection problem to minimize total delay and global model, and solve it using a Lyapunov-based heuristic algorithm. Simulation results demonstrate that the proposed method outperforms baseline schemes in terms of total delay and model accuracy.**

***Index Terms*—Internet of Vehicles, LEO satellite-assisted vehicular networks, decentralized federated learning, total delay.**

I. INTRODUCTION

WITH the rapid development of intelligent vehicles, the Internet of Vehicles (IoV) has emerged as a key technology for intelligent transportation systems (ITS) applications, such as autonomous driving, in-vehicle monitoring, and traffic flow prediction. In particular, for autonomous driving, intelligent vehicles perform real-time obstacle detection and avoidance using machine learning (ML) models, which require training on large volumes of data (*e.g.,* LiDAR point clouds and camera images) [1].

However, transmitting massive amounts of raw data from vehicles to the central server for ML model training leads to privacy concerns and communication overhead. Therefore, federated learning (FL) has emerged as a promising solution, where vehicles locally train models and upload the model parameters to a parameter server (PS), instead of raw data [1], [2]. Nevertheless, the mobility of vehicles or urban obstructions often leads to unstable communication links between vehicles and terrestrial infrastructures. Moreover, in remote areas such as rural or disaster-stricken regions where terrestrial infrastructures are limited or damaged, communication links are unavailable [3]. These intermittent communication links result in transmission failures in uploading model parameters from vehicles to the PS, degrading the model accuracy [4].

While unmanned aerial vehicles (UAVs) have been utilized to collect the model parameters to enhance communication reliability in vehicular networks, their short operational time due to limited battery capacity causes frequent recharging and redeployment, increasing network management overhead [5]. Therefore, low Earth orbit (LEO) satellite-assisted decentralized FL (DFL) has been proposed, where LEO satellites with global coverage and seamless connectivity act as PSs to aggregate model parameters from vehicles. In LEO satellite-assisted DFL, satellites collaboratively train a global model by exchanging and aggregating models via inter-satellite links (ISLs) [6], [7]. However, previous studies have mainly focused on scenarios where satellites perform only model aggregation task as PSs. To assist intelligent vehicles, the satellites not only aggregate model parameters, but also simultaneously process ML-based inference tasks, such as extracting map features from satellite imagery for high-definition (HD) map generation [3]. As the computational load of the required inference tasks differs across regions, some satellites take longer to complete the aggregation task due to the waiting delay caused by executing inference tasks. The different total delays to complete one round of FL at each satellite slow down the convergence and potentially reduce the accuracy of the global model, degrading the quality of services for intelligent vehicles, such as untimely driver alerts for object avoidance or inaccurate traffic flow prediction [2].

Therefore, in this paper, we propose a delay-aware DFL framework in LEO satellite-assisted vehicular networks, considering the different waiting delays caused by simultaneously executed inference tasks at each satellite. Specifically, we analyze the waiting delay by modeling the queue backlog of the inference tasks that contribute to the total delay. We then formulate a satellite selection problem to optimize total delay and model accuracy under long-term queuing constraints. As the formulated problem is a nonlinear 0-1 knapsack problem, which is widely known to be NP-hard [8], we design a heuristic-based satellite selection algorithm by adopting the Lyapunov optimization method to ensure long-term queue stability. The simulation results show that the proposed algorithm outperforms the baseline schemes in terms of total delay and model accuracy.

II. SYSTEM MODEL

A. Network Model

We consider LEO satellite-assisted vehicular networks consisting of a LEO constellation with LEO satellites $m \in \mathcal{M}$, intelligent vehicles $k \in \mathcal{K}$, and a domain controller (DC) located on an upper-layer satellite (*i.e.,* medium-earth orbit) [9], as illustrated in Fig. 1. The vehicles equipped with

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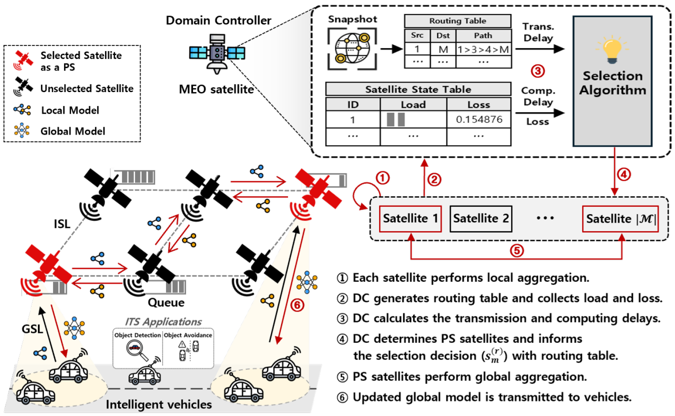


Fig. 1. Overall procedure of the proposed framework.

steerable antennas on the rooftop communicate directly with LEO satellites via ground-satellite links (GSLs) [10], while the satellites are interconnected via inter-satellite links (ISLs). Referring to [11], GSLs and ISLs can be established using the radio resource control (RRC) and Xn protocols, respectively.

At the beginning of each FL round, $r \in \mathcal{R}$, the proposed framework operates as follows: (1) Each satellite aggregates local models from vehicles within its coverage via GSL. (2) The DC captures the topology of the LEO constellation as a snapshot and computes the ISL routing paths between satellites using the shortest path algorithm (*e.g.,* Dijkstra's algorithm) [7]. The computed paths are stored in a routing table and transmitted to each satellite. The routing tables are updated whenever the constellation moves from one snapshot to another. (3) The DC collects the computational load ($Q_{m,i}^{(r)}$) for each satellite and then calculates the delay ($\tau_m^{\text{glb},(r)}$). (4) Based on the computed delay, the DC selects multiple satellites as PSs using the proposed selection algorithm (detailed in Section III) and informs all satellites of the selection result ($s_m^{(r)}$). (5) The selected satellites collaboratively perform global aggregation by forwarding their locally aggregated models along the multi-hop ISL path using the routing table disseminated in step (2). When the selected satellite ($s_m^{(r)} = 1$) receives locally aggregated models, it aggregates them with its own model; otherwise ($s_m^{(r)} = 0$), it relays them to the next satellite. The global aggregation for round r terminates once all selected satellites have exchanged their models. (6) Subsequently, each satellite disseminates the updated global model to vehicles.

B. LEO satellite-assisted DFL Process

The LEO satellite-assisted DFL proceeds through local training, local aggregation, and global aggregation. The details are as follows.

1) Local training: Let $w_k^{(r)}$ denote the local model parameters of vehicle k in round r. The vehicle k trains the local model based on its local dataset $\mathcal{D}_k$ during L epochs, which loss function is $F_k(w) = \frac{1}{|\mathcal{D}_k|}\sum_{x\in\mathcal{D}_k} f(x;w)$, where $f(x;w)$ denotes the loss on x-th data sample. The local model parameters are updated as $w_k^{(r)} = w_k^{(r-1)} - \eta_k \nabla F_k(w_k^{(r-1)})$, where η_k is the learning rate of the vehicle k, and $\nabla F_k(w_k^{(r-1)})$ is the gradient computed with the sampled mini-batches by stochastic gradient descent (SGD).

2) Local aggregation: After local training, each vehicle uploads its updated local model to a connected satellite. Then, each satellite m performs local aggregation (*e.g.,* FedAvg) as $w_m^{(r)} = \frac{1}{|\mathcal{D}_m|}\sum_{k\in\mathcal{K}_m} |\mathcal{D}_k| w_k^{(r)}$, where $\mathcal{K}_m$ is a set of vehicles within the coverage of satellite m, and $|\mathcal{D}_m| = \sum_{k\in\mathcal{K}_m} |\mathcal{D}_k|$ is the total amount of training data of $\mathcal{K}_m$. Then, the loss of the aggregated model at satellite m is

$$F_m(w) = \frac{1}{|\mathcal{D}_m|}\sum_{k\in\mathcal{K}_m} |\mathcal{D}_k| F_k(w). \tag{1}$$

3) Global aggregation: Let $s_m^{(r)}$ denote a binary variable, where $s_m^{(r)} = 1$ if satellite m is selected for model aggregation in round r, and 0 otherwise. After local aggregation, each satellite m performs global aggregation by sharing local models with selected satellites as $w^{(r)} = \frac{1}{|\mathcal{D}|}\sum_{m\in\mathcal{M}} s_m^{(r)} |\mathcal{D}_m| w_m^{(r)}$, where $|\mathcal{D}| = \sum_{m\in\mathcal{M}} s_m^{(r)}|\mathcal{D}_m|$. The global loss function can be written as

$$F(w) = \frac{1}{|\mathcal{D}|}\sum_{m\in\mathcal{M}} s_m^{(r)} |\mathcal{D}_m| F_m(w). \tag{2}$$

C. Communication Model

Let $\mathcal{K}_m$ denote a set of vehicles within the link-of-sight (LoS) coverage of satellite m, which consists of devices satisfying the minimum elevation angle α_{el} condition as $\angle(\mathbf{v}_m - \mathbf{v}_k, \mathbf{v}_k) \leq \frac{\pi}{2} - \alpha_{\text{el}}$, where $\angle(\cdot)$ is the angle between two vectors, $\mathbf{v}_m$ and $\mathbf{v}_k$ are the Earth-centered Earth-fixed (ECEF) position vectors of the vehicle k and the satellite m, respectively. To satisfy the minimum elevation angle, the distance between the vehicle and the satellite, denoted by $d_{k,m}$, should satisfy the following constraint: $d_{k,m} \leq \sqrt{r^2 - r_e^2 + r_e^2 \sin^2 \alpha_{\text{el}}} - r_e \sin \alpha_{\text{el}}$, where r and r_e denote the orbital radius of the satellite and the radius of the Earth, respectively. We assume that the vehicle position is static, as the vehicle mobility is negligible compared to the satellite movement [12]. According to Shannon's theorem, the transmission rate from vehicle k to satellite m is given by $R_{k,m}^{\text{GSL}} = B_k \log_2\left(1 + \frac{p_k |G_{k,m}|^2}{B_k N_0}\right)$, where B_k is the channel bandwidth, p_k is the transmission power, N_0 is the noise power spectral density. The channel gain between vehicle k and satellite m is modeled as $G_{k,m} = \beta_0/(d_{k,m})^\gamma g$, where γ is the path loss exponent, β_0 is the channel gain, and g is the Rayleigh fading parameter. The transmission delay for vehicle k to upload the local model to satellite m is given by:

$$\tau_{k,m}^{\text{GSL},(r)} = \frac{|w_k^{(r)}|}{R_{k,m}^{\text{GSL}}}, \tag{3}$$

where $|w_k^{(r)}|$ denotes the size (in bits) of local model.

The set of neighboring satellites, $\mathcal{N}_m$, which can establish a valid ISL with satellite m, is defined by the following LoS condition: $d_{m,n} \leq \sqrt{(h_n + r_e)^2 - r_e^2} + \sqrt{(h_m + r_e)^2 - r_e^2}\}$, where $d_{m,n}$ is the Euclidean distance between satellites m and n, h_m and h_n are their altitudes, respectively. The transmission rate for ISL from satellite m to n can be written as $R_{m,n}^{\text{ISL}} = B_m \log_2\left(1 + \frac{p_m G_m^{\text{Tx}} G_n^{\text{Rx}}}{B_m \zeta_{m,n} N_0}\right)$, where B_m is the bandwidth, p_m is the transmit power, G_m^{Tx} and G_n^{Rx} are the transmitter and receiver antenna gains, respectively, and $\zeta_{m,n}$ is the free-space path loss. The one-hop transmission delay over ISL is given by

$$\tau_{m\to n}^{\text{ISL},(r)} = \frac{|w_m^{(r)}|}{R_{m,n}^{\text{ISL}}}, \quad n \in \mathcal{N}_m, \tag{4}$$

where $|w_m^{(r)}|$ is the size of the locally aggregated model from the satellite m.

D. Computation Model

Let c_k denote the CPU cycles required to process a bit of data for local training at vehicle k. Let f_k (in cycles/sec) denote the CPU frequency of vehicle k. The local training delay for vehicle k to train the model in round r is

$$\tau_k^{\text{train},(r)} = \frac{c_k L |\mathcal{D}_k^{(r)}|}{f_k}. \tag{5}$$

Let $c_{m,0}$ denote the CPU cycles required to process a bit of data for local and global aggregation at satellite m. The aggregation delays for satellite m to perform local and global aggregations in round r can be calculated as

$$\tau_m^{\text{agg},(r)} = \frac{c_{m,0}|w_m^{(r)}|}{f_m}, \quad \tau_m^{\text{agg}',(r)} = \frac{c_{m,0}|w^{(r)}|}{f_m}, \tag{6}$$

where f_m is the CPU frequency of the satellite m.

We consider that satellites perform a set of inference tasks, where $i \in \mathcal{I}$ denotes a task type. Let $c_{m,i}$ denote the CPU cycles required to process a bit of data for inference task i at satellite m. We assume that the arrival of inference tasks at satellite m follows a Poisson process [13]. For each task $i \in \mathcal{I}$, the number of tasks arriving in round r, denoted by $N_{m,i}^{(r)}$, follows a Poisson distribution with an average arrival rate of $\lambda_{m,i}$. The total computational load newly arriving at satellite m is $\Lambda_{m,i}^{(r)} = N_{m,i}^{(r)} \cdot \delta_i$, where δ_i is the size of an inference task i. Let $Q_{m,i}^{(r)}$ denote the queue backlog (in bits) of the inference tasks queued at satellite m, which evolves as

$$Q_{m,i}^{(r+1)} = \max\left\{0, Q_{m,i}^{(r)} - \mu_{m,i}^{(r)}\right\} + \Lambda_{m,i}^{(r)}, \tag{7}$$

where $\mu_{m,i}^{(r)}$ is the amount of computational load processed at round r. The waiting delay caused by processing the previously accumulated inference tasks in the queue is [14]:

$$\tau_m^{\text{wait},(r)} = \frac{\sum_{i\in\mathcal{I}} c_{m,i} Q_{m,i}^{(r)}}{f_m}. \tag{8}$$

E. Total Delay Model

The delay for satellite m to complete the local aggregation is determined by the vehicle with the slowest computation and transmission capabilities, and is defined as

$$\tau_m^{\text{loc},(r)} = \max_{k\in\mathcal{K}_m}\left(\tau_k^{\text{train},(r)} + \tau_{k,m}^{\text{GSL},(r)}\right) + \tau_m^{\text{agg},(r)}. \tag{9}$$

After local aggregation, the selected satellites exchange their aggregated models for global aggregation. The transmission delay from satellite m to another satellite m' along the shortest path $p_{m,m'}^{(r)}$ provided by the DC at each snapshot, as mentioned in Section II-A, is expressed as

$$\tau_{m\to m'}^{\text{path},(r)} = \sum_{(u,v)\in p_{m,m'}^{(r)}} \tau_{u\to v}^{\text{ISL},(r)}. \tag{10}$$

For satellite m to perform global aggregation, it should receive the models from all other selected satellites. The total transmission delay is determined by the slowest transmission link from any other selected satellite m' to m, which can be expressed as

$$\tau_m^{\text{trans},(r)} = \max_{m'\in\mathcal{M}\setminus\{m\}}\left(s_{m'}^{(r)} \cdot \tau_{m'\to m}^{\text{path},(r)}\right). \tag{11}$$

Then, the delay for satellite m to complete global aggregation can be obtained as

$$\tau_m^{\text{glb},(r)} = \max\{\tau_m^{\text{trans},(r)}, \tau_m^{\text{wait},(r)}\} + \tau_m^{\text{agg}',(r)}. \tag{12}$$

Finally, the total delay required to complete global model training in round r is defined as

$$\tau^{(r)} = \max_{m\in\mathcal{M}}\left\{s_m^{(r)} \cdot \left(\tau_m^{\text{loc},(r)} + \tau_m^{\text{glb},(r)}\right)\right\}. \tag{13}$$

We formulate a satellite selection problem aimed at minimizing both the loss of the final global model and the total delay as follows:

$$\min_{\{s_m^{(r)}\}} \quad \alpha F(w) + (1-\alpha)\sum_{r=1}^{|\mathcal{R}|} \tau^{(r)} \tag{14a}$$

$$\text{s.t.} \quad s_m^{(r)} \in \{0,1\}, \quad \forall m \in \mathcal{M},\ \forall r \in \mathcal{R}, \tag{14b}$$

$$\sum_{m\in\mathcal{M}} s_m^{(r)} \le |\mathcal{M}|, \quad \forall r \in \mathcal{R}, \tag{14c}$$

$$\lim_{R\to\infty} \frac{1}{R}\sum_{r\in\mathcal{R}}\sum_{i\in\mathcal{I}} \frac{Q_{m,i}^{(r)}}{\overline{\Lambda_{m,i}^{(r)}}} \le \delta, \quad \forall m \in \mathcal{M}, \tag{14d}$$

where $\alpha \in [0,1]$ is a weighting factor that balances the learning performance and total delay over $|\mathcal{R}|$. The constraint (14d) ensures that the long-term queuing delay at each satellite does not exceed the maximum queuing delay δ. The time-averaged task arrival rate is defined as $\overline{\Lambda_{m,i}^{(r)}} = \frac{1}{r}\sum_{r'=0}^{r-1} \Lambda_{m,i}^{(r')}$.

III. Delay-aware satellite selection

Dynamic changes in the collected training data caused by vehicle mobility and varying traffic conditions across regions lead to uncertainty in the loss $F(w)$, which makes it difficult to solve the problem formulated in Eq. (14a). Specifically, the relationship between $s_m^{(r)}$ and $w^{(r)}$ cannot be expressed in a simple closed form because $w^{(r)}$ evolves with the selection decisions $s_m^{(r)}$ across rounds, depending on the amount of collected data and its distribution among the vehicles. Therefore, we transform $F(w)$, by adopting a combination of age of update and training loss methods to measure global model convergence as $C^{(r)} = -\frac{\left(\sum_{m=1}^{M} s_m^{(r-1)} \mathcal{D}_m O_m^{(r-1)} \mathcal{L}_m^{(r-1)}\right)^{1-\beta}}{1-\beta}$, where $\mathcal{L}_m^{(r)} = \frac{1}{|\mathcal{K}_m|}\sum_{k\in\mathcal{K}_m} F_k(w)$ denotes the average loss of vehicles served by satellite m, $O_m^{(r-1)}$ is the age of model update, and $\beta \in (0,1)$ is a sensitivity parameter [15].

We then leverage Lyapunov optimization to develop a satellite selection algorithm that ensures long-term queue stability. The Lyapunov function is $L(Q^{(r)}) = \frac{1}{2}\sum_{m\in\mathcal{M}}(\sum_{i\in\mathcal{I}} Q_{m,i}^{(r)})^2$. Moreover, the one-step Lyapunov drift is derived as $\Delta L(Q^{(r)}) = \mathbb{E}\left[L(Q^{(r+1)}) - L(Q^{(r)}) \mid Q^{(r)}\right]$. According to Lyapunov optimization theory, the drift-plus-penalty is

$$\Delta L(Q^{(r)}) + V\mathbb{E}[\alpha C^{(r)} + (1-\alpha)\tau^{(r)}], \tag{15}$$

where $V > 0$ denotes the control parameter. The drift-plus-penalty term is upper-bounded as $\Delta_V L(Q^{(r)}) \leq B + V\mathbb{E}[\alpha C^{(r)} + (1-\alpha)\tau^{(r)}] + \sum_{m\in\mathcal{M}}\sum_{i\in\mathcal{I}}\mathbb{E}[Q_{m,i}^{(r)}(\Lambda_{m,i}^{(r)} - \mu_{m,i}^{(r)})]$, where B is a nonnegative constant. Following Lyapunov optimization theory to minimize the upper bound of the drift-plus-penalty, Eq. (14a) is transformed as

$$\min_{\{s_m^{(r)}\}} V\mathbb{E}[\alpha C^{(r)} + (1-\alpha)\tau^{(r)}] - \sum_{m\in\mathcal{M}}\sum_{i\in\mathcal{I}}\mathbb{E}[Q_{m,i}^{(r)}\mu_{m,i}^{(r)}]. \tag{16}$$

Since this problem is NP-hard, as it is equivalent to a nonlinear 0-1 knapsack problem [8], we propose a heuristic algorithm that selects satellites as PSs based on a score that reflects how much the total delay decreases and the learning performance increases in the global model when they are selected. The DC computes the score for every satellite $m \in \mathcal{M}$ in each round r as

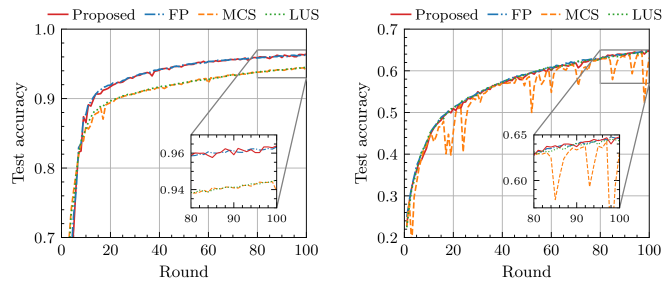
$$\psi_m^{(r)} = V\left(\alpha C_m^{(r)} - (1-\alpha)\tau_m^{est,(r)}\right) - \sum_{i\in\mathcal{I}} Q_{m,i}^{(r)}\Delta\mu_{m,i}^{(r)}, \tag{17}$$

where $C_m^{(r)} = \mathcal{D}_m\mathcal{O}_m^{r-1}\mathcal{L}_m^{r-1}$ is the training gain of satellite m, $\tau_m^{est,(r)} = \tau_m^{\text{loc},(r)} + \tau_m^{\text{wait},(r)}$ is an estimated delay, and $\Delta\mu_{m,i}^{(r)}$ represents the service rate degradation for inference tasks. After calculating the scores for all satellites, the DC computes a threshold, which is the average score: $\hat{\psi}^{(r)} = \frac{1}{|\mathcal{M}|}\sum_{m\in\mathcal{M}}\psi_m^{(r)}$. The DC sets $s_m^{(r)}$ to 1 if $\psi_m^{(r)} > \hat{\psi}^{(r)}$, and 0 otherwise. Finally, the DC obtains the set of selected satellites in round r, denoted by $\mathcal{S}^{(r)} = \{m \mid s_m^{(r)} = 1, \forall m \in \mathcal{M}\}$.

IV. Performance Evaluation

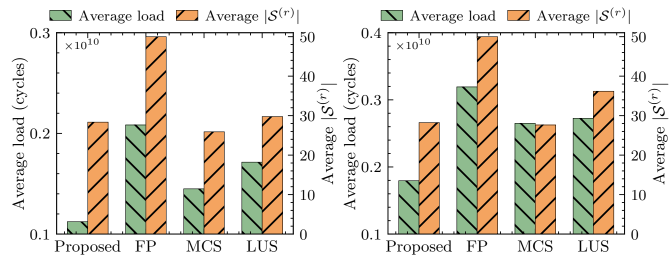
A. Simulation Settings

The simulation of the LEO satellite constellation is implemented using the Starlink two-line element (TLE) dataset, which provides time-varying satellite positions [16]. We consider 50 satellites uniformly distributed across ten orbital planes at an altitude of 550 km, each moving at a velocity of 7.56 km/s. The vehicles are randomly distributed across urban and rural areas, with densities of 40 and 20 per km^2, respectively, and move at speeds ranging from 40 to 120 km/h [3]. We use the MNIST dataset with a multi-layer perceptron (MLP) model [7], and additionally train a convolutional neural network (CNN) model using the CIFAR-10 dataset for evaluating performance on image data used in intelligent vehicle services, such as obstacle detection [6]. The channel bandwidth is set to 20 MHz for both $B_{k,m}$ and $B_{m,n}$. The distribution of local datasets $|\mathcal{D}_m|$ across the vehicles follows a Dirichlet distribution with a parameter of 0.5 [17]. The FL training is conducted over $|\mathcal{R}| = 100$ rounds, where each vehicle performs local training for $L = 5$ epochs with the learning rate of $\eta_k = 0.01$. Referring to [3], [7], [15], the remaining parameters are set as $f_k = 1$ GHz, $f_m = 10$ GHz, $c_k = c_{m,0} = c_{m,i} = 120$ cycles/bit, $\delta_i = 5$ MB.



(a) MNIST (b) CIFAR-10

Fig. 2. Test accuracy comparison of different schemes.



(a) MNIST (b) CIFAR-10

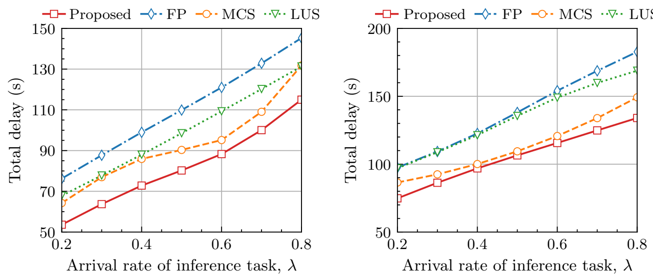
Fig. 3. Average computational load of selected satellites and the average number of selected satellites.

We evaluate the proposed scheme in terms of the total delay and test accuracy compared with three benchmark methods: 1) Full Participation (FP): all satellites act as PSs in DFL [6]; 2) Monte Carlo-based Selection (MCS): satellites are selected uniformly via Monte Carlo sampling [18]; 3) Load-Unaware Selection (LUS): the selection decision is made based on transmission, local training, and aggregation delays [19].

B. Simulation Results

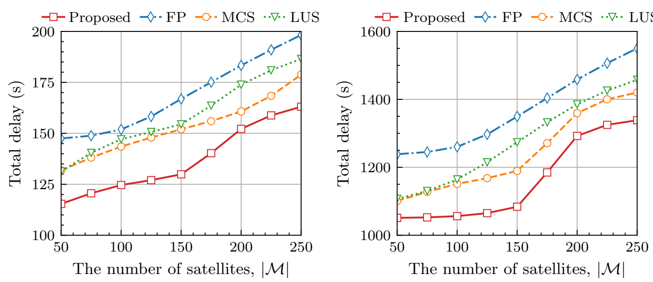
Fig. 2 illustrates the evolution of the test accuracy of the global model over the training rounds for two different datasets: (a) MNIST and (b) CIFAR-10. We observe that the proposed, FP, and LUS schemes converge within 65 rounds on the MNIST dataset and 85 rounds on the CIFAR-10 dataset. In contrast, the MCS scheme does not converge on the CIFAR-10 dataset, because the randomly selected PSs, without considering the loss value of the satellites, cause the aggregation of poorly trained local models. The proposed and FP schemes show the highest test accuracy, while FP has the longest total delay, as shown in Figs. 4 and 5, and involves twice as many participating satellites compared to the proposed scheme, as shown in Fig. 3, indicating that the proposed scheme results in lower communication overhead.

Fig. 3 shows the average number of selected satellites $\mathbb{E}[|\mathcal{S}^{(r)}|] = \frac{1}{|\mathcal{R}|}\sum_{r=1}^{|\mathcal{R}|}|\mathcal{S}^{(r)}|$ and the average computational load, defined as the mean queue backlog over the selected satellites, $\mathbb{E}[Q^{(r)}] = \frac{1}{|\mathcal{S}^{(r)}|}\sum_{m\in\mathcal{M}} s_m^{(r)}\sum_{i\in\mathcal{I}} Q_{m,i}^{(r)}$. We observe that the proposed scheme selects fewer PSs than the other



(a) MNIST (b) CIFAR-10

Fig. 4. Total delay with respect to the arrival rate of inference tasks.



(a) MNIST (b) CIFAR-10

Fig. 5. Total delay with respect to the number of satellites.

baseline schemes, resulting in a lower average computational load for the selected satellites while still achieving the highest accuracy, as shown in Fig. 2. This is because the proposed scheme selects satellites based on not only the queue backlog but also the loss value of the locally aggregated model on each satellite. MCS chooses the least number of satellites, but the selected satellites have higher computational loads than the proposed scheme, resulting in the increased total delay, as shown in Figs. 4 and 5.

Fig. 4 illustrates the total delay, which represents the total time taken to train the global model after 100 rounds, as the arrival rate of inference tasks increases. As the arrival rate increases from 0.2 to 0.8, the total delay also increases for all schemes due to the higher number of inference tasks queued at the satellites. The proposed scheme achieves the lowest total delay by 24.87%, 9.47%, and 19.32% on average compared with FP, MCS, and LUS, respectively. This improvement is achieved because the proposed scheme selects satellites by considering both the waiting delay caused by the inference tasks and the transmission delay between the satellites. LUS has a higher total delay than the proposed scheme because it does not consider the waiting delay at the satellites or the transmission delay between satellites.

Fig. 5 shows the total delay as the number of satellites increases. The proposed scheme reduces the total delay by 17.1%, 8.5%, and 11.2% compared to FP, MCS, and LUS, respectively. This is because the proposed scheme selects satellites by considering the transmission delay when sharing the model parameters between satellites during global aggregation.

V. Conclusion

In this paper, we propose a delay-aware DFL framework in LEO satellite-assisted vehicular networks, considering the waiting delay caused by simultaneously executed inference and FL aggregation tasks at each satellite. We formulate a satellite selection problem to reduce the total delay and global model loss, and develop a Lyapunov-based heuristic algorithm. Simulation results show that the proposed scheme outperforms the benchmark schemes.

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